turgor

Illinois analysis + code

One soybean stress index, validated on two continents

Turgor turns soybean physiology (when drought and heat hit flowering and seed fill) into a weekly county stress map and expected loss, from satellite and weather data only. Built in Brazil; tested in Illinois.

Why

Lenders, crop insurers and grain buyers learn the size of a soybean shortfall late. In Illinois the loss below trend averaged **US$ 23.5/acre a year** (2001–2025), **US$ 57/acre** in a bad year and **US$ 104/acre in 2012** (at US$ 12.55/bu). July–August rainfall is the usual weather signal; it can miss stress at seed fill.

What

Weekly, June to October: crop stage from MODIS NDVI (corn–soybean pixels, USDA Cropland Data Layer), water deficit, heat above 30 °C and their interaction from NASA POWER, weighted 3× at flowering and seed fill. Same formula and weights as in Brazil: nothing was tuned on Illinois.

Illinois evidence (102 counties, 2001–2023 backtest; 2,194 county-seasons)

	Turgor	Rainfall Jul–Aug	Palmer Aug	NDVI only
AUC, loss > 10%	**0.81**	0.75	0.69	0.68
Share of losses missed (basis risk)	**0.47**	0.57	0.54	0.72
Statewide-year correlation	**0.60**	0.44	0.34	0.44

Honest limits: the gain over rainfall is not statistically proven in Illinois (better in 78–85% of resampled season sets), and on the 2024–25 holdout (17 losses) rainfall did better. **2003:** July–August rain looked normal, the index ranked the season 78th percentile of stress; yields ended −17%. **15 July 2012:** 99 of 102 counties rated severe; they ended −17%. **Brazil** (780 municipalities): AUC 0.86 vs 0.78 for rainfall, gain proven, 8–12 weeks before harvest.

For whom

Reinsurers and parametric providers working in Brazil and the US; lenders and crop insurers watching a soybean book in season; grain buyers forecasting supply. Pricing to test: per portfolio per year, or US$ 0.25–2 per acre per season.

Looking for

Conversations with lenders, insurers and grain buyers about how they track crop risk today. **Collaborators:** remote sensing and data science; agricultural finance and insurance. **Advisors:** crop insurance, farm credit, crop modelling.

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